

Design and dynamics of two-qubit gates with motional states of electrons on helium

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Systems of individual electrons electrostatically trapped on condensed-noble-gas surfaces have recently attracted considerable interest as potential platforms for quantum computing. The electrons serve as charge qubits in the system, and the purity of the noble-gas surface protects the relevant quantum properties of each electron. Previous work has indicated that manipulation of a confining double-well potential for electrons on superfluid helium can generate entanglement suitable for two-qubit gate operations. In this work, we incorporate a time-dependent tuning of the potential shape to further explore operation of two-qubit gates with the superfluid helium system. Through numerical time evolution of the closed system (without decoherence), we show that control-induced errors can be minimized to allow for fast high-fidelity two-qubit gates. In particular, we simulate operation of the $\sqrt{i}$ SWAP and CZ gates and obtain estimated fidelities of 0.999 and 0.996 with execution times of 2.9 and 9.4 ns, respectively. Furthermore, we examine the stability of these gate fidelities under nonideal execution conditions, which reveals new properties to consider in the device design. Finally, we reflect on the impact of screening and decoherence on our results. The methodology presented here enables future efforts to isolate control-induced effects from environmental noise, which is an important step towards the realization of high-fidelity two-qubit gates with electrons on helium.

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I. INTRODUCTION

Recent advances in trapping single electrons on condensed-noble-gas surfaces, such as liquid helium [1,2] or solid neon [3–6], have positioned these systems as promising platforms for studying light-matter interactions and for applications in quantum computing. In particular, bound electron states on liquid ^4He surfaces offer significant potential due to their precise spatial control and shuttling capabilities [7], enabled by the defect- and impurity-free nature of the pristine helium substrate. The electron's spin degree of freedom is predicted to exhibit long coherence times, owing to the absence of magnetic impurities in the helium dielectric environment and the extremely weak spin-orbit interaction [8]. Additionally, both quantized out-of-plane and in-plane motional states have been proposed as qubit systems, where the spatial charge states can be tuned via applied dc potentials [9,10]. In a system of two or more electrons, quantum correlations, which are a central component in the quantum computing paradigm, can be induced by the Coulomb interaction between the charged electrons [11,12]. Therefore, a detailed understanding and control of the quantum degrees of freedom of both single

electrons and small electron ensembles is of significant interest for quantum technologies.

The detection of individual electrons and few-electron ensembles on liquid helium has been facilitated by ongoing developments in circuit quantum electrodynamics, which serves as a foundational framework for superconducting qubit architectures and for high-sensitivity charge detection in semiconductor quantum-dot systems [13,14]. Experiments have demonstrated coupling between microwave cavity photons and the cyclotron motion of a macroscopically large number of electrons on the surface of liquid helium [15]. Newer studies have shown that the motion of a single electron within an in-plane confining potential also can be coupled to the field of a microwave cavity [1,2], with strong coupling recently realized experimentally [16]. Under certain conditions, the quantized electron motion can be represented as a two-level system [16]. The coupled electron-cavity system can then be described by a Jaynes-Cummings-type Hamiltonian, which enables the use of standard quantum nondemolition measurement techniques to infer the quantized motional state of the electron [10]. Single-qubit gates can be implemented by applying voltage drives at the qubit transition frequency via gate electrodes [13].

Several schemes have been proposed for implementing two-qubit gates, which rely on the Coulomb interaction between two electrons on the liquid-helium surface [11,12,17,18]. Many of these proposed schemes rely on

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simplifications, where smooth confining potentials are typically approximated as harmonic-oscillator-type potentials near the equilibrium point. However, in general the trapping potentials are inherently nonlinear, leading to unequally spaced energy levels, and thus more complex dynamics in the system. This is especially important in developing two-qubit gates, where the electron frequencies need to be tuned by adjusting the gate electrode potentials. This tuning can cause unwanted excitations into higher excited states, limiting gate fidelities and posing challenges for achieving high-fidelity quantum operations.

The shape and placement of the gate electrodes under the helium surface are especially important because they determine how the electrons' in-plane charge states are formed and controlled. Electrodes with dimensions on the order of several hundreds of nanometers are typically used to trap a small number of electrons on the helium surface. The in-plane electrostatic confinement can then be controlled by adjusting the voltages applied to the gate electrodes [1,2]. When several electrodes are involved, the trapping potential becomes an increasingly complex function of the applied voltages. We previously studied how entanglement between two electrons can be created through the tuning of the confining potential and how nonlinearity can be deliberately engineered and controlled to enforce specific types of correlations between the two electrons, which is a prerequisite for implementing two-qubit gates in systems of electrons on helium [11].

Here we expand upon our earlier work by proposing two-qubit gate protocols which utilize the entanglement generation from our previous publication [11] and an alternative approach for entanglement generation. Moreover, we use a time-dependent full configuration-interaction-inspired method [19] for distinguishable particles to simulate the dynamics of the protocol and proceed to quantify the gate performance with the average gate fidelity [20]. This allows us to maximize the fidelity with respect to the two parameters of the gate protocol, that is, the hold and ramp times. We specifically target the $\sqrt{i}$ SWAP and controlled-Z (CZ) gates, for which we obtain fidelities of 0.999 and 0.996 with experimentally achievable execution times of 2.9 and 9.4 ns, respectively. Finally, we examine the sensitivity of these gate fidelities to perturbations in the hold and ramp times. Our results can aid the experimental realization of two-qubit gates with systems of electrons on helium.

The paper is arranged as follows. Section II A describes the device used in our simulations, and the operation of the proposed gate protocol is detailed in Sec. II B. The numerical method used to optimize the parameters of the protocol is outlined in Sec. II C. In Secs. III A and III B we present the results for the $\sqrt{i}$ SWAP and CZ gates, respectively. The sensitivity of the gates to ramp and hold time deviations is analyzed in Sec. III C, and the impact of screening and decoherence is discussed in Sec. III D. We summarize, present our conclusions, and discuss the outlook for future work in Sec. IV. Some more in depth analyses are provided in the Appendices.

II. DEVICE AND OPERATION

Electrons near the surface of liquid helium polarize the helium atoms, resulting in a weak attractive force that pulls

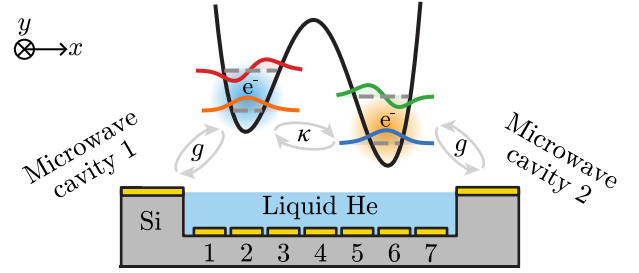


FIG. 1. Schematic microdevice, in which two electrons are trapped on the surface of a liquid-helium basin in an electrostatic double-well potential created by electrodes 1–7. The two-qubit gates are driven by the Coulomb interaction κ between the two electrons, while single-qubit gates and readout are performed through the electron-cavity interaction g .

them toward the surface. At the helium-vacuum interface, the electrons experience hard-core repulsion with a potential barrier of around 1 eV, which prevents them from penetrating the liquid. The resulting out-of-plane interaction with the helium surface leads to a quantized Rydberg-like energy spectrum, and at sufficiently low temperatures $T < 2$ K, the electrons only occupy the ground state. The electron's in-plane motion is controlled by a confining potential created by electrodes positioned around and beneath the helium surface. This work builds upon our previous publication [11], utilizing the same device geometry and entanglement generation scheme reported in that study. Here we expand our analysis of the system by proposing two-qubit gate protocols and simulating the dynamics of the system, in order to predict and optimize the fidelities of two-qubit gates.

A. Device description

The device geometry is schematically depicted in Fig. 1. It consists of seven control gate electrodes designed to create an electrostatic double-well potential on the helium surface, which confines two electrons in separate potential wells. The electrodes are 200 nm wide, spaced by 200 nm and positioned beneath a 500-nm-deep layer of liquid helium. This geometry was chosen to create an in-plane motional quantization axis along the x direction, with energy gaps in the frequency range of 5–15 GHz. These states are decoupled from motional states along the y direction at approximately six times larger frequencies, thereby allowing us to ignore the states along the y axis [11]. Each electron is dipole coupled to its own microwave cavity, enabling both qubit state readout and operation of single-qubit gates. The Coulomb interaction between the two electrons, which are separated by a distance of approximately $1.5 \mu\text{m}$ for the well configurations used in this study, provides a mechanism through which we can introduce the necessary entanglement to operate two-qubit gates.

The Hamiltonian of the system is given by

$$\hat{H} = \sum_{i=1}^2 \left[-\frac{1}{2} \frac{d^2}{dx_i^2} + v(x_i) \right] + u(x_1, x_2), \quad (1)$$

where $u(x_1, x_2) = \kappa / \sqrt{(x_1 - x_2)^2 + \epsilon^2}$ is the Coulomb interaction with a strength parameter $\kappa = 2326$ determined by the

choice of energy unit [11], and $\epsilon = 0.01$ is a shielding parameter introduced to remove the singularity at $x_1 = x_2$ [21]. Due to the proximity of the underlying electrodes, the Coulomb interaction is reduced by screening effects. However, we employ an unscreened interaction in our analysis, which provides an upper bound for the interaction strength. We discuss the implications of the screening in Sec. III D. The electrostatic potential from the electrodes beneath the condensed-helium layer is given by

$$v(x) = -\sum_{k=1}^7 \alpha_k(x) V_k \equiv -\alpha(x) \cdot V, \quad (2)$$

where V_k is the voltage applied to electrode k and $\alpha_k(x)$ is a dimensionless function quantifying the relative contribution of the electrode k to the electrostatic potential at the helium surface, which is calculated using the finite element method [11].

B. Gate operation

In order to introduce dynamics to the system, we parametrize the voltage vector in Eq. (2) through voltage functions $V(\lambda)$. These functions are constructed such that the system can alternate between different configurations, i.e., shapes of the electrostatic double-well potential, each with distinct properties. Following Ref. [11], we distinguish between three configurations. In configuration I the system is in an idle state, where the effective interaction leading to entanglement between the two electrons is minimized, thus enabling readout of the qubit states as well as single-qubit gate operation. Configuration II is designed to favor an interaction suitable for swap-type gates, whereas the interaction in configuration III is tailored to facilitate for CZ gates. Incorporating the voltage vectors of such configurations, two different voltage functions will be explored in this work. The system properties for these functions are presented in Ref. [11] and Appendix A, respectively. The first voltage function, which we will denote by V^β , reads

$$V^\beta(\lambda) = (1 - \lambda)V_I^\beta + \lambda V_{III}^\beta, \quad (3)$$

where V_I^β and V_{III}^β are the constant voltage vectors from our previous work, which were optimized for opposite-sign anharmonicities $\beta^L = -\beta^R$ in the two wells [11,22]. Opposite-sign anharmonicity was however shown to be insufficient to suppress unwanted ZZ coupling in the entanglement driven by the Coulomb interaction [11]. Therefore, we also present results for a second voltage function, denoted by V^ζ , given by

$$V^\zeta(\lambda) = (1 - \lambda)V_I^\zeta + \lambda V_{II,III}^\zeta, \quad (4)$$

where the last term is set to either V_{II}^ζ or V_{III}^ζ , depending on the desired type of entanglement. For this voltage function, the main goal in the optimization of the constant voltage vectors V_I^ζ , V_{II}^ζ , and V_{III}^ζ was to directly minimize the ZZ coupling, quantified by

$$\zeta = E_4 - E_2 - E_1 + E_0, \quad (5)$$

where E_i is the i th eigenenergy of the full system. This optimization is described in more detail in Appendix A, together with the resulting energy spectra and von Neumann entropies.

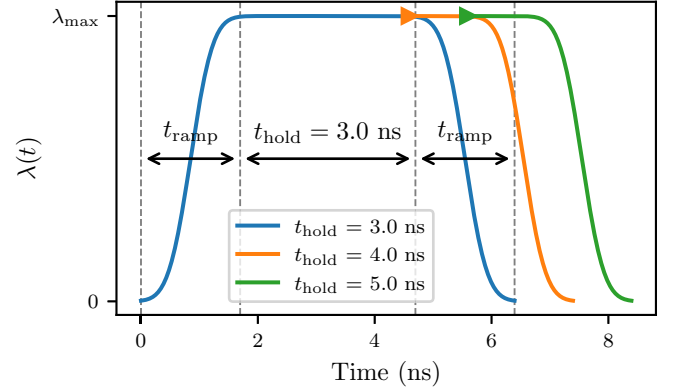


FIG. 2. The blue line shows the shape of $\lambda(t)$ with $t_{\text{ramp}} = 0.3 \times 4\sqrt{2} \text{ ns} \approx 1.7 \text{ ns}$ and $t_{\text{hold}} = 3.0 \text{ ns}$. The orange and green lines illustrate how calculations with longer hold times, in this example 4.0 and 5.0 ns, respectively, can be warm-started using previous calculations, thus reducing the computational cost.

Two-qubit gate operations are enabled by introducing a time dependence in the parametrization of the voltage functions, $V(\lambda) = V(\lambda(t))$. The time-dependent parameter $\lambda(t)$ should induce a smooth transition from the idle configuration I to one of the entangling configurations (II or III) and then back, as time progresses [23]. Two-qubit gates can then be realized by driving the system into an entangling configuration, where the system is kept for a duration t_{hold} to generate entanglement, before returning to configuration I, where the electronic states can be regarded as qubit states. However, at intermediate timescales, where the transition between the configurations is neither fully diabatic nor adiabatic, entanglement is also generated during the transition itself. To accurately capture the full dynamics of the gate protocol, we employ a numerical method, described in Sec. II C, which provides an exact solution to the time-dependent problem across all relevant timescales.

Inspired by [24], the explicit function shape of the time-dependent parameter $\lambda(t)$ was chosen to be

$$\lambda(t) = \frac{\lambda_{\text{max}}}{2} \left[\text{erf} \left(\frac{t - \frac{1}{2}t_{\text{ramp}}}{\sqrt{2}\sigma} \right) - \text{erf} \left(\frac{t - t_{\text{gate}} + \frac{1}{2}t_{\text{ramp}}}{\sqrt{2}\sigma} \right) \right], \quad (6)$$

where λ_{max} is the maximum value the function assumes, $t_{\text{gate}} = t_{\text{hold}} + 2t_{\text{ramp}}$ is the total gate time, t_{hold} is the time for which the system is kept in an entangling configuration (II or III), and t_{ramp} is how quickly the system transitions between the idle configuration I and the desired entangling configuration. The function is illustrated in Fig. 2. Throughout this work, we let $t_{\text{ramp}} = 4\sqrt{2}\sigma$ [24].

For both voltage functions, configuration I is located at $\lambda(t) = 0$, i.e., when $t = 0$ or $t > t_{\text{gate}}$, while the entangling configurations are realized when $t \in (t_{\text{ramp}}, t_{\text{ramp}} + t_{\text{hold}})$. For the first voltage function V^β , the type of entanglement, i.e., configuration II or III, is decided by setting λ_{max} to $\lambda_{II} \equiv 0.46$ or $\lambda_{III} \equiv 1$, respectively [11]. For the second voltage function

V^ζ , $\lambda_{\max}$ should be 1 for both types of entanglement, and the configuration is instead determined by choosing either V_{II}^ζ or V_{III}^ζ in the second term of Eq. (4). In order to utilize the entanglement of configurations II and III to drive specific two-qubit gates, what remains is then to determine the hold and ramp times that give the best gate performance.

C. Optimization of gate fidelity

We employ a grid search approach to optimize the ramp and hold times for maximum gate fidelity. The efficiency of the search is improved by noting that for each fixed value of the ramp time, the calculation of a new hold time can be warm-started using the previous calculation, as illustrated in Fig. 2.

In our numerical method, we assume that the potential wells are deep enough for the two trapped electrons to effectively become distinguishable [11]. This assumption allows us to expand any two-body state $|\Psi(t)\rangle$ in a tensor product basis of sinc-discrete variable representation (DVR) [25] functions,

$$|\Psi(t)\rangle = \sum_{\alpha\beta} C_{\alpha\beta}(t) |\chi_\alpha^L \chi_\beta^R\rangle, \quad (7)$$

where χ_α^L and χ_β^R represent two distinct sets of sinc-DVR functions defined on discrete grid points in the left and right wells, respectively. Using the matrix elements derived in Appendix B of [11], we can then write the effect of $\hat{H}(t)$ on a state $|\Psi(t)\rangle$ as

$$\begin{aligned} [\hat{H}(t)|\Psi(t)]_{\gamma\delta} &\equiv \langle \chi_\gamma^L \chi_\delta^R | \hat{H}(t) | \Psi(t) \rangle \\ &= \sum_{\alpha\beta} C_{\alpha\beta}(t) [\delta_{\delta\beta} (t_{\gamma\alpha}^L + v_\alpha^L(\lambda(t))) \\ &\quad + \delta_{\gamma\alpha} (t_{\delta\beta}^R + v_\beta^R(\lambda(t))) + \delta_{\gamma\alpha} \delta_{\delta\beta} u_{\alpha\beta}], \end{aligned} \quad (8)$$

where the subscripts on the left-hand side denote elements in the coefficient matrix of the state $\hat{H}(t)|\Psi(t)\rangle$ for a tensor product expansion like (7). With this expression established, the lowest-energy eigenstates can be determined iteratively using the Davidson method [26,27]. We compute the six lowest eigenstates at $t=0$, i.e., when the system is in configuration I. These eigenstates correspond to the computational basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|02\rangle$, $|11\rangle$, and $|20\rangle$, respectively [11]. Among them, the four relevant qubit states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, i.e., the eigenstates $|\Phi_0\rangle$, $|\Phi_1\rangle$, $|\Phi_2\rangle$, and $|\Phi_4\rangle$, are propagated in time using the Crank-Nicolson propagator [20]. We denote the propagated state which initially is equal to eigenstate $|\Phi_n\rangle$ by $|\Psi_n(t)\rangle$. The system is evolved for a time $t_{\text{gate}} = t_{\text{hold}} + 2t_{\text{ramp}}$ with an integration step size of $\Delta t = (\omega_{\text{qubit}}/2\pi)^{-1}/100 = 0.001$ ns, suitable for dynamics on the qubit energy scale of $\omega_{\text{qubit}}/2\pi \sim 10$ GHz, which is typical for our system [11]. After the time evolution, the 4×4 matrix U describing the qubit dynamics is given by the overlap matrix between the four two-qubit states at $t=0$ and $t=t_{\text{gate}}$,

$$U_{ij} = \langle \Phi_{n_i} | \Psi_{n_j}(t_{\text{gate}}) \rangle, \quad (9)$$

where $n_k = k$ for $k \in \{0, 1, 2\}$ while $n_3 = 4$ (omitting state 3 from the matrix since it corresponds to the unwanted $|02\rangle$ state). Single-qubit rotations $R(\theta_L, \theta_R) = R_z(\theta_L) \otimes R_z(\theta_R)$

[28] are then applied, yielding an overall two-qubit gate matrix $G(\theta_L, \theta_R) = R(\theta_L, \theta_R)U$.

We quantify the performance of the gate $G(\theta_L, \theta_R)$ by the average gate fidelity [29]

$$F = [\text{Tr}(MM^\dagger) + |\text{Tr}(M)|^2]/20. \quad (10)$$

Here $M \equiv U_0^\dagger G(\theta_L, \theta_R)$, where U_0 is the gate matrix describing an ideal operation of the targeted two-qubit gate. The single-qubit rotation angles θ_L^{opt} and θ_R^{opt} that maximize the fidelity are found using a standard optimizer [30], and the final optimized two-qubit gate is then given by $G(\theta_L^{\text{opt}}, \theta_R^{\text{opt}})$.

III. RESULTS AND DISCUSSION

In this section, we present results from the simulation of two different two-qubit gates, the $\sqrt{i\text{SWAP}}$ gate and the CZ gate, for both of the voltage functions described in Sec. II B. The gate accuracy up to single-qubit rotations is quantified using the fidelity measure from Eq. (10) and the gate time is reported.

A. $\sqrt{i\text{SWAP}}$ gate

The first two-qubit gate we strive to achieve is the $\sqrt{i\text{SWAP}}$ gate

$$\sqrt{i\text{SWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & i/\sqrt{2} & 0 \\ 0 & i/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (11)$$

For a perfect $\sqrt{i\text{SWAP}}$ gate, a pure $|01\rangle$ or $|10\rangle$ state should be swapped into a maximally entangled state with equal probability of $|01\rangle$ and $|10\rangle$. We can thus define a swap error to measure the deviation from the ideal probabilities, where 0 indicates perfect agreement and 1 represents the maximum error, according to

$$\epsilon_{\text{swap}} = \sum_{i,j=1}^2 |0.5 - |U_{ij}||^2/2, \quad (12)$$

where U_{ij} is given by Eq. (9). We conduct a sweep over hold times from 0 to 5 ns with steps of $\Delta t_{\text{hold}} = 0.1$ ns and over ramp times from $t_{\text{ramp}} = 0.05 \times 4\sqrt{2}$ ns ≈ 0.28 ns to $t_{\text{ramp}} = 0.5 \times 4\sqrt{2}$ ns ≈ 2.83 ns with steps of $\Delta t_{\text{ramp}} = 0.01 \times 4\sqrt{2}$ ns ≈ 0.06 ns. The obtained swap error ϵ_{swap} for the two different voltage functions is depicted in Fig. 3(a) for V^β and in Fig. 3(b) for V^ζ , with the corresponding gate fidelities shown in Figs. 3(c) and 3(d), respectively.

Comparing the four plots in Fig. 3, we find that ϵ_{swap} behaves similarly across the two voltage functions. However, identifying a meaningful resemblance between ϵ_{swap} and the fidelity is challenging, demonstrating that ϵ_{swap} is a bad predictor of the gate fidelity and thus that the relative phases of the states could be more important than their amplitudes. The structure of the fidelity differs substantially between the two voltage functions, with the fidelity decaying significantly with increased ramp time for V^β . The reason for this decay is mainly the poor suppression of the ZZ coupling provided by the opposite-sign anharmonicities [11]. Both cases reveal

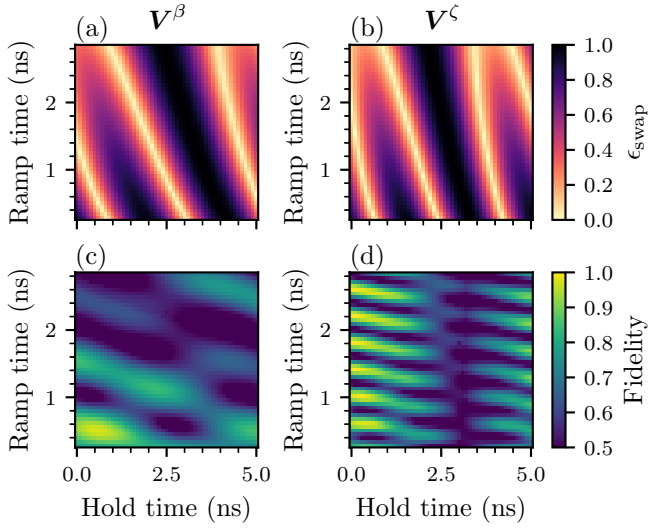


FIG. 3. Grid search for an $\sqrt{i\text{SWAP}}$ gate with step sizes $\Delta t_{\text{ramp}} = 0.01 \times 4\sqrt{2} \text{ ns} \approx 0.06 \text{ ns}$ and $\Delta t_{\text{hold}} = 0.1 \text{ ns}$. The swap error ϵ_{swap} is shown for (a) V^β and (b) V^ζ and the corresponding gate fidelities of the two voltage functions are displayed in (c) and (d), respectively. The maximum fidelity achieved for V^β is $F = 0.971$, obtained at $t_{\text{ramp}} = 0.51 \text{ ns}$ and $t_{\text{hold}} = 0.6 \text{ ns}$, and for V^ζ the maximum fidelity is $F = 0.999$, realized with $t_{\text{ramp}} = 1.41 \text{ ns}$ and $t_{\text{hold}} = 0.1 \text{ ns}$.

oscillatory behavior as a function of ramp time. This oscillation is linked to the energy difference between the first and second excited states of the system, which in turn establishes the phase difference of their respective matrix elements. This phase difference is crucial for the performance of the $\sqrt{i\text{SWAP}}$ gate. A more detailed analysis of the features in the fidelity plots is provided in Appendix B.

The maximum fidelity achieved by voltage function V^β is $F = 0.971$, with $t_{\text{ramp}} = 0.51 \text{ ns}$ and $t_{\text{hold}} = 0.6 \text{ ns}$, corresponding to a total gate time of 1.6 ns . A more accurate gate is acquired with voltage function V^ζ , reaching an optimal fidelity of $F = 0.999$ for $t_{\text{ramp}} = 1.41 \text{ ns}$ and $t_{\text{hold}} = 0.1 \text{ ns}$, corresponding to a total gate time of 2.9 ns . This gate operates more slowly than the gate associated with V^β ; however, it should be noted that if the total execution time is limited to that of the V^β gate (durations around 1.6 ns), both voltage functions achieve the same fidelity.

The matrices of the best gates we obtained can be written as

$$G_{\sqrt{i\text{SWAP}}}^\beta = \begin{pmatrix} 1.00 & 0 & 0 & 0 \\ 0 & \sqrt{0.51}e^{i0.12\pi} & \sqrt{0.49}e^{i0.61\pi} & 0 \\ 0 & \sqrt{0.49}e^{i0.63\pi} & \sqrt{0.51}e^{i0.11\pi} & 0 \\ 0 & 0 & 0 & 1.00 \end{pmatrix}$$

for V^β and

$$G_{\sqrt{i\text{SWAP}}}^\zeta = \begin{pmatrix} 1.00 & 0 & 0 & 0 \\ 0 & \sqrt{0.53} & \sqrt{0.47}e^{i0.49\pi} & 0 \\ 0 & \sqrt{0.47}e^{i0.50\pi} & \sqrt{0.53} & 0 \\ 0 & 0 & 0 & 1.00 \end{pmatrix}$$

for V^ζ . Despite the higher fidelity displayed by the V^ζ gate, it is evident that the amplitudes of the $|01\rangle \leftrightarrow |10\rangle$ swap are worse than for the V^β gate, further strengthening the claim that the relative phases are the most important factors for the $\sqrt{i\text{SWAP}}$ gate fidelity. Nevertheless, the gate fidelity is influenced by both the amplitude and phase of the elements associated with the swap, as well as the amplitude and phase of the G_{33} element. As the ZZ coupling decreases, the variation in the latter element reduces, thus increasing the probability that all matrix elements obtain desirable values for a certain combination of t_{hold} and t_{ramp} , which is why the V^ζ gate achieves better fidelity.

B. CZ gate

The search for a high-fidelity CZ gate is performed in a similar fashion to the case of the $\sqrt{i\text{SWAP}}$ gate described in the preceding section, but with the voltage functions modified to arrive at configuration III after time $t = t_{\text{ramp}}$, as described in Sec. II B. Expressed as a matrix, the CZ gate reads

$$\text{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (13)$$

Thus, the gate produces a π phase shift on the $|11\rangle$ state while preserving all other states. We aim to achieve this phase shift by strengthening the interactions among the $|02\rangle$, $|11\rangle$, and $|20\rangle$ states. However, this approach entails a risk of population leakage from $|11\rangle$ to the nonqubit states $|02\rangle$ and $|20\rangle$. We therefore define a leakage error

$$\epsilon_{\text{leak}} = 1 - |U_{33}|^2, \quad (14)$$

where $U_{33} = \langle \Phi_4 | \Psi_4(t_{\text{gate}}) \rangle$ [see Eq. (9)]. We then perform a sweep over t_{hold} and t_{ramp} , utilizing the same step sizes as in the $\sqrt{i\text{SWAP}}$ case, but with the search area increased such that the ramp time ranges from $t_{\text{ramp}} = 0.05 \times 4\sqrt{2} \text{ ns} \approx 0.28 \text{ ns}$ to $t_{\text{ramp}} = 0.7 \times 4\sqrt{2} \text{ ns} \approx 3.96 \text{ ns}$ for V^β and the hold time ranges from 0 to 10 ns for V^ζ . Figure 4 visualizes the results from the sweep, with the leakage error ϵ_{leak} shown in Fig. 4(a) for V^β and in Fig. 4(b) for V^ζ . The corresponding gate fidelities are depicted in Figs. 4(c) and 4(d), respectively.

In contrast to the amplitude-based swap error of the $\sqrt{i\text{SWAP}}$ gate, we observe that the leakage error ϵ_{leak} , which is also based solely on the amplitudes, matches the CZ gate fidelity quite well. Some additional structure emerges for the V^β fidelity, shown in Fig. 4(c), due to unwanted interactions between the $|01\rangle$ and $|10\rangle$ states during the transition from configuration I to configuration III. This interaction reduces the fidelity unless the hold time compensates for the oscillations between these states, as detailed in Appendix B. The other key difference between the two voltage functions is the oscillation frequency of both the fidelity and ϵ_{leak} relative to hold time, which is about twice as large for V^β . (Note the different ranges on the x axes in Fig. 4.) This frequency results from the dynamics among the three excited states $|\Psi_3(t)\rangle$, $|\Psi_4(t)\rangle$, and $|\Psi_5(t)\rangle$. The same dynamics are also responsible for the fidelity not attaining its maximum until the second period of the oscillation, and only at certain ramp

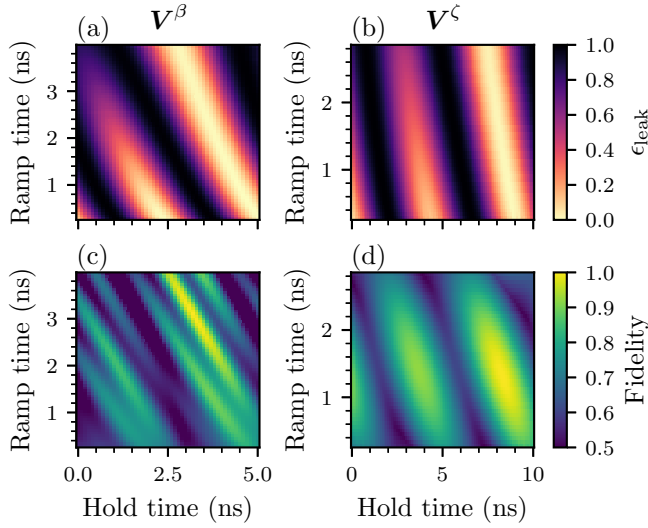


FIG. 4. Grid search for a CZ gate with step sizes $\Delta t_{\text{ramp}} = 0.01 \times 4\sqrt{2} \text{ ns} \approx 0.06 \text{ ns}$ and $\Delta t_{\text{hold}} = 0.1 \text{ ns}$. Note that the ranges of the axes are different between the two voltage functions. The leakage error ϵ_{leak} is shown for (a) V^β and (b) V^ζ and the corresponding gate fidelities are displayed in (c) and (d), respectively. The maximum achieved fidelity is $F = 0.996$ for both voltage functions. With V^β this value is obtained at $t_{\text{ramp}} = 3.11 \text{ ns}$ and $t_{\text{hold}} = 3.2 \text{ ns}$, whereas it is given by $t_{\text{ramp}} = 1.30 \text{ ns}$ and $t_{\text{hold}} = 8.3 \text{ ns}$ with V^ζ .

times. However, the intricacies of the time evolution make it challenging to directly attribute the behavior of these dynamics to specific characteristics of the different voltage functions.

The best gate fidelity achieved with voltage function V^β is $F = 0.996$, which is obtained for $t_{\text{ramp}} = 3.11 \text{ ns}$ and $t_{\text{hold}} = 3.2 \text{ ns}$, yielding a total gate time of 9.4 ns . Voltage function V^ζ produces the same fidelity $F = 0.996$; however, it is obtained at $t_{\text{ramp}} = 1.30 \text{ ns}$ and $t_{\text{hold}} = 8.3 \text{ ns}$, resulting in a slower execution time of 10.9 ns . The reduced speed is a result of the lower frequency discussed above. In terms of matrices, the best gate we achieve with V^β can be expressed as¹

$$G_{\text{CZ}}^\beta = \begin{pmatrix} 1.00 & 0 & 0 & 0 \\ 0 & 1.00e^{-i0.01\pi} & 0.02e^{-i0.29\pi} & 0 \\ 0 & 0.02e^{-i0.70\pi} & 1.00e^{-i0.01\pi} & 0 \\ 0 & 0 & 0 & 0.99e^{-i1.00\pi} \end{pmatrix}$$

and with V^ζ can be written as¹

$$G_{\text{CZ}}^\zeta = \begin{pmatrix} 1.00 & 0 & 0 & 0 \\ 0 & 1.00 & 0.02e^{i0.67\pi} & 0 \\ 0 & 0.03e^{i0.56\pi} & 1.00 & 0 \\ 0 & 0 & 0 & 0.99e^{-i1.00\pi} \end{pmatrix}.$$

We note that both gate matrices have some unwanted off-diagonal elements, as well as a minor leakage of the G_{33} element. Additionally, V^β has a small phase error on the diagonal G_{11} and G_{22} elements. Still, all these errors are small, hence the high fidelities.

¹The two center column vectors only appear to not be normalized because the values are rounded.

C. Sensitivity to ramp and hold deviations

Finally, we conduct a more thorough investigation into the stability of the gate fidelity as the hold and ramp times deviate from the ideal values identified in the previous two sections. The fidelities of the gates in a window of $\pm 0.1 \text{ ns}$ around the optimal values of t_{hold} and t_{ramp} are presented in Figs. 5(a)–5(d), with corresponding one-dimensional cross sections shown in Figs. 5(e)–5(h). In these calculations, the resolution is 0.01 ns for both hold and ramp times.

In the case of the $\sqrt{i}$ SWAP gate, shown in Fig. 5 for V^β [Figs. 5(a) and 5(e)] and for V^ζ [Figs. 5(b) and 5(f)], the gate performance remains relatively stable with nonideal hold times. However, it is very sensitive to deviations in ramp time, particularly for the V^ζ voltage function. These properties are a result of the phase obtained through the $|01\rangle \leftrightarrow |10\rangle$ swap (see Appendix B for details). During the ramp-up from configuration I to configuration II, the average energy difference between the $|\Psi_1(t)\rangle$ and $|\Psi_2(t)\rangle$ states is 0.94 GHz for V^β and 2.51 GHz for V^ζ . In contrast, the same energy difference during the hold and ramp-down stages of the gate operation is only about 0.1 GHz . The accumulated phase differences of the corresponding matrix elements are determined by integrals of the energy difference over time. Consequently, the relatively large energy difference during the ramp-up stage causes the phase difference to predominantly be determined by the ramp time. This explains the observed sensitivity of gate fidelity with respect to ramp time deviations, while the relatively small energy difference during the hold stage explains the stability to deviations in hold time.

The smaller energy difference, and thus better stability, of the gate associated with V^β can be attributed to two factors. First, the energy difference in configuration I is smaller, and second, the energies of the states cross during the time evolution, resulting in $|\Psi_1(t)\rangle$ having a higher energy than $|\Psi_2(t)\rangle$ after the ramp-up is finished, thereby reducing the average energy difference. Both these factors should be incorporated in future optimizations of electrode voltages V_{II} in order to improve the stability of the $\sqrt{i}$ SWAP gate.

The CZ gate with the V^β voltage function exhibits about the same sensitivity for both parameters [see Figs. 5(c) and 5(g)], with a slightly larger dependence on the hold time. Similarly to before, this sensitivity mainly arises from interactions between the $|\Psi_1(t)\rangle$ and $|\Psi_2(t)\rangle$ states, which are entirely unwanted for this gate. However, in this case the energy difference between these states is roughly the same during all three stages of the gate execution, thus explaining the similar sensitivity with respect to both t_{ramp} and t_{hold} .

All three gates discussed thus far might be problematic to implement experimentally due to the relatively fast drop in fidelity if the precision by which the gate parameters can be tuned is insufficient. The last gate, CZ implemented with the V^ζ voltage function, is on the contrary highly stable, mainly because interactions between $|\Psi_1(t)\rangle$ and $|\Psi_2(t)\rangle$ remain suppressed throughout the whole gate protocol. As seen in Figs. 5(d) and 5(h), the drop in fidelity is less than 1% for ramp and hold time deviations of 0.1 ns , which should allow decent gate performance with current control electronics [31].

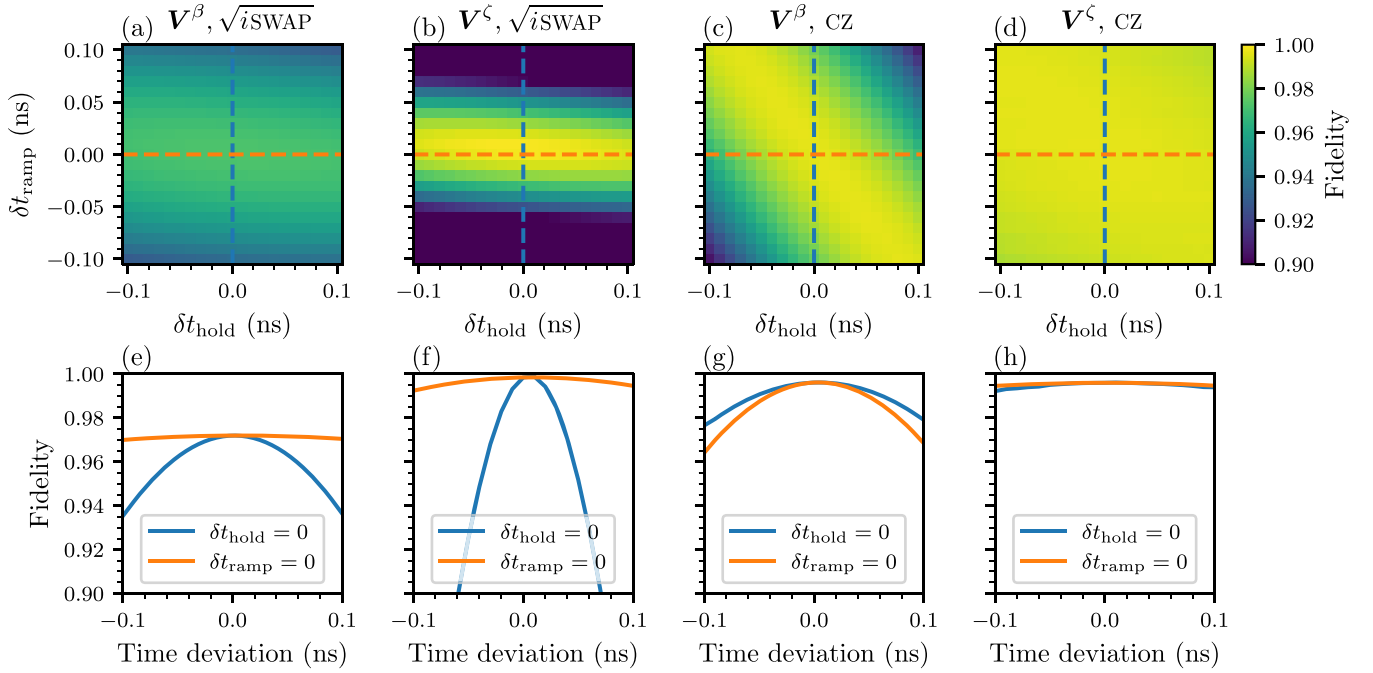


FIG. 5. Gate fidelity for small deviations δt_{ramp} and δt_{hold} from the optimal ramp and hold times. The change in fidelity is presented for the $\sqrt{i}\text{SWAP}$ gate using (a) V^β and (b) V^ζ . For the CZ gate, the same results are shown with (c) V^β and (d) V^ζ . Corresponding one-dimensional cross sections are presented in (e)–(h). Note that the ramp time is crucial for the $\sqrt{i}\text{SWAP}$ gate's performance.

D. Screening and decoherence

We conclude our discussion by addressing two physical effects omitted from our current model that would impact the performance of an experimental realization: screening of the Coulomb interaction between the electrons and decoherence of the electron motional states.

The proximity of the electrons to the metallic electrodes beneath the helium layer gives rise to a screening of the Coulomb interaction by free charges in the electrodes. To estimate the impact of this screening on the two-qubit gates, it is crucial to isolate its effect on the two-particle coupling strength that leads to entanglement.

As detailed in Appendix D, we model the screening using the method of image charges. An expansion of the interaction potential reveals that the screening has two distinct contributions. The constant and first-order terms in the expansion lead to shifts in the total energy and equilibrium positions. While significant, these shifts primarily affect the qubit frequencies and the resonance conditions for the gates. In our protocol, these effects can be compensated for by modifying the external potential through reoptimization of the voltage functions $V(\lambda)$ to restore the necessary avoided crossings between energy levels.

The second contribution arises from the curvature of the screened interaction potential, which couples the displacements of the two electrons. It is this entangling coupling that drives the gate dynamics and determines the gate speed. In Appendix D we derive the screening factor η for this coupling strength. For our device geometry, with an electron separation of $d \approx 1.5 \mu\text{m}$ and helium depth $h = 500 \text{ nm}$, we calculate a screening factor of $\eta \approx 0.69$, which implies an increase in gate time by a factor of roughly $1/\eta \approx 1.45$. This moderate

increase in gate time suggests that the high fidelities reported in this work can be maintained even in the presence of screening, provided the above-mentioned retuning of the voltage functions is implemented.

Furthermore, the overall reduction of the Coulomb interaction strength due to screening is beneficial for our voltage optimization, as it effectively localizes the regions in voltage space where different motional states of the electrons couple. This localization improves the isolation of desired interactions from unwanted ones in the different configurations. Consequently, the fidelities reported in this work, where decoherence effects are not considered, would increase.

In a realistic experimental scenario, the performance of the two-qubit gates will however be limited by decoherence of the electron motional states. Recent experiments [16] indicate that the charge qubit lifetime is predominantly limited by dephasing, with measured rates on the order of 60 MHz. This is comparable to the two-qubit gate rates obtained in this work, implying that further suppression of dephasing will be necessary to achieve high-fidelity gates. Although the underlying source of this dephasing is not yet fully established, two primary mechanisms have been proposed: fluctuations from background charges localized on the helium surface near the electron and coupling to a bath of thermally excited ripplons (surface capillary waves). In the former case, the fluctuating charges effectively modulate the electrostatic potential, causing random variations in the qubit frequency resulting in motional dephasing. This mechanism may be eliminated through device and cell design improvements aimed towards more control of the generation and trapping of electrons on the helium surface.

Regarding the latter mechanism, the presence of thermally excited ripplons modulates the energies of the electron

motional states, which leads to dephasing. (See Appendix E for an analysis of the coupling Hamiltonian.) The electron coupling to the ripplonic bath can be quantified by a dimensionless parameter $\mathcal{C}$, following Ref. [32]. By varying the external pressing field $E_{\perp}$, the coupling parameter can be tuned from a strong-coupling regime $\mathcal{C} \gg 1$ to a weak-coupling regime $\mathcal{C} \ll 1$. In the strong-coupling regime the electron absorption spectrum has a Gaussian shape with a characteristic width $\gamma \propto \sqrt{\mathcal{C}}$ [33]. For temperatures $T = 10$ mK and pressing fields $E_{\perp} = 1000$ V/cm, the parameter $\mathcal{C}$ is about 10 and the coupling to ripplons is strong. This gives a linewidth of the absorption spectrum of $\gamma/2\pi \approx 100$ MHz, and in this regime coherent Rabi oscillations between electron motional states are not expected. Lower pressing fields are required to get to the weak-coupling regime. For $E_{\perp} = 10$ V/cm the linewidth is $\gamma/2\pi \approx 10$ MHz; however, in this regime a narrow zero-ripplon line (ZRL) appears atop the broader Gaussian background. This is analogous to the zero-phonon lines in the spectra of color centers [34,35]. Under these weak-coupling conditions the ZRL becomes the most prominent feature in the spectrum. The corrections to the Rabi dynamics are expected to be small in this regime [33]. Therefore, achieving high two-qubit gate fidelities requires operation in the low-pressing-field regime, necessitating careful tuning of the voltages applied to the underlying electrodes and adding a new parameter to consider in the optimization process.

A complete treatment of ripplon-induced dephasing requires a more rigorous analysis based on the Lindblad master equation, providing insight into the underlying physics across different parameter regimes [36]. The ability to tune the coupling strength between electron motional states and the ripplonic bath over an energy scale comparable to the Coulomb interaction between two electrons provides a promising platform for investigating the interplay between dephasing and collective effects in coupled-electron systems [37,38].

IV. CONCLUSION

In summary, we have shown that the entangling configurations from our previous work [11] can be utilized to achieve high-fidelity two-qubit gates, with the non-Clifford $\sqrt{i}$ SWAP gate and the CZ gate as concrete examples. We have also shown that a direct minimization of the ZZ coupling due to Coulomb interaction improves the fidelity for the $\sqrt{i}$ SWAP gate compared to optimization for opposite-sign anharmonicities. On the other hand, it reduces the speed of the CZ gate due to the dynamics among the higher excited states. The exact cause of the reduced speed remains to be determined.

Our investigation of the stability of the gate fidelity reveals that even changes in ramp time smaller than a tenth of a nanosecond can negatively impact gate fidelity. For the $\sqrt{i}$ SWAP gate, this impact can be reduced by minimizing the average energy difference between the first and second propagated eigenstates $|\Psi_1(t)\rangle$ and $|\Psi_2(t)\rangle$, through the ramp-up stage of the gate. Hence, it is desirable to optimize the electrostatic double-well configurations such that these two states switch order between the idle and entangling stages of the gate. In the case of the CZ gate, we manage to achieve a stable

gate performance with the V^{ζ} voltage function due to a strong suppression of swap-type interactions.

Although the scope of this study was limited to the motional states of the electrons, we emphasize the importance of our results also for qubits of electrons on helium utilizing spin degrees of freedom. Given that current spin-state readout methods rely on spin-to-charge conversion techniques [10,39], the thorough understanding of charge dynamics and electrostatic control presented here is essential not only for charge-based architectures but also as a foundation for spin-based approaches.

The theoretical insights presented in this work are particularly timely given recent experimental milestones. Strong coupling between the motional states of a single electron on helium in a quantum dot and microwave photons in a cavity has recently been demonstrated experimentally [16]. It has also been shown that such quantum dots can be reliably loaded with two electrons, making the study of two-electron dynamics experimentally accessible. We anticipate that these developments, guided by the methodology presented here to minimize gate errors in a closed system, will enable future efforts to effectively isolate control-induced effects from environmental noise. Achieving this isolation is a prerequisite for understanding and mitigating decoherence, which is the next crucial step towards the first experimental realization of high-fidelity two-qubit gates with electrons on helium.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this article are openly available [40].

APPENDIX A: OPTIMIZATION OF THE V^{ζ} VOLTAGES

The optimization of voltages for the V^{ζ} voltage function was primarily concentrated on directly minimizing the ZZ interaction, $\zeta = E_4 - E_2 - E_1 + E_0$, for all configurations. In addition, the following loss terms were used when optimizing for specific configurations.

- (1) *Configuration I* consists of one term to punish frequencies outside the 5–15 GHz operating range of the microwave cavities used for single-qubit operations and one term to punish qubit detuning smaller than 3 GHz. These terms were weighted by a factor 10^{-2} relative to the ζ term.
- (2) *Configuration II* involves one term to minimize $(E_2 - E_1)^2$ to encourage an avoided crossing between the first and second excited states, which was weighted by a factor 10^{-4} relative to the ζ term. Additionally, two terms to punish energy differences between the third, fourth,

TABLE I. Electrode voltage vectors $\mathbf{V}_I^\zeta$, $\mathbf{V}_{II}^\zeta$, and $\mathbf{V}_{III}^\zeta$ used in the $\mathbf{V}^\zeta$ voltage function defined in Eq. (4).

V_k	V_I^ζ (mV)	V_{II}^ζ (mV)	V_{III}^ζ (mV)
V_1	389.50	388.68	388.17
V_2	200.70	206.69	194.01
V_3	400.36	404.88	401.87
V_4	-290.61	-288.37	-289.10
V_5	398.59	401.04	398.95
V_6	200.15	192.98	198.82
V_7	381.40	382.57	382.44

and fifth excited states smaller than 1.5 GHz were used in order to keep these energy levels separated, hence suppressing the interaction between them. These two terms were weighted by a factor 10^{-2} relative to the ζ term.

- (3) *Configuration III* contains three terms for the triple degeneracy point between $|\Phi_3\rangle$, $|\Phi_4\rangle$, and $|\Phi_5\rangle$. The first two of these minimize the energy gaps $\Delta E_{54}^2 \equiv (E_5 - E_4)^2$ and $\Delta E_{43}^2 \equiv (E_4 - E_3)^2$ to enforce the degeneracy point. The third term minimizes $(\Delta E_{54} - \Delta E_{43})^2$ to symmetrize the two avoided crossings in the degeneracy point. These three terms had the same weight as the ζ term. Furthermore, the loss function included one term to punish energy differences between the first and

second excited states below 1.5 GHz in order to keep these energy levels separated and thereby suppress the swap-type interaction. This term had a weight of 10^4 relative to the ζ term.

The weights of the different loss terms above were chosen heuristically by trial and error. We refer to our previous work [11] for a more elaborate description of the optimization procedure. Optimization of these loss functions yielded the electrode voltage vectors $\mathbf{V}_I^\zeta$, $\mathbf{V}_{II}^\zeta$, and $\mathbf{V}_{III}^\zeta$ used in the $\mathbf{V}^\zeta$ voltage function defined in Eq. (4). Explicit values of the electrode voltages are reported in Table I. The energy spectra and von Neumann entropies of the two resulting voltage functions (one for configuration II and one for configuration III) are presented in Fig. 6. The energy gap between $|\Phi_1\rangle$ and $|\Phi_2\rangle$ for configuration II is 297 MHz, while the gap between $|\Phi_3\rangle$ and $|\Phi_5\rangle$ for configuration III is 565 MHz. A similar figure for the $\mathbf{V}^\beta$ voltage function can be found in Ref. [11], where the energy gaps in the avoided crossing points have values of 225 and 377 MHz, respectively.

APPENDIX B: ELEMENTWISE GATE MATRIX ANALYSIS

In order to better understand the patterns that emerge in the gate fidelity, we will analyze the amplitude and phase of each matrix element separately in this Appendix. We choose

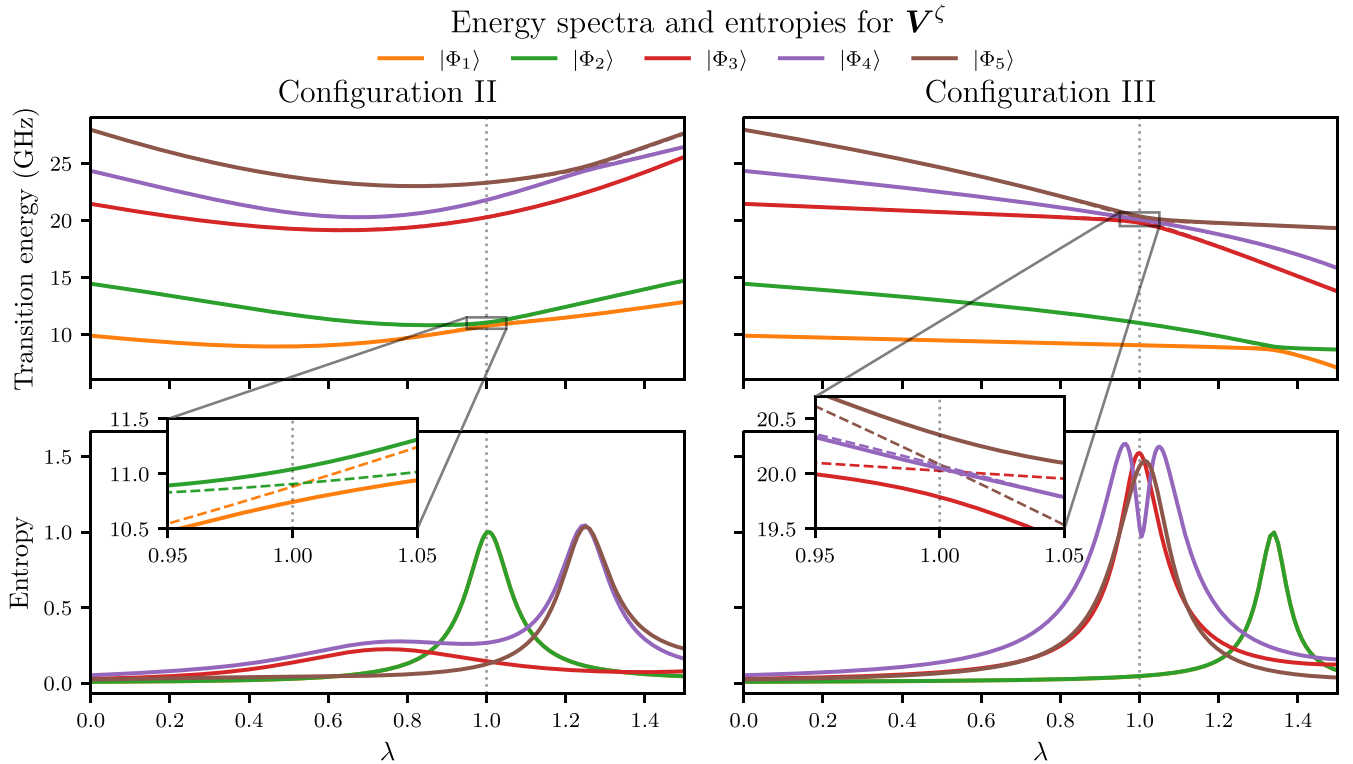


FIG. 6. Transition energies from the ground state (top row) and von Neumann entropies (bottom row) of the five lowest excited energy eigenstates for $\mathbf{V}^\zeta(\lambda) = (1 - \lambda)\mathbf{V}_I^\zeta + \lambda\mathbf{V}_{II,III}^\zeta$, with $\mathbf{V}_{II,III}^\zeta = \mathbf{V}_{II}^\zeta$ (left) and $\mathbf{V}_{II,III}^\zeta = \mathbf{V}_{III}^\zeta$ (right). Dashed lines in the top plots represent the transition energies of the corresponding noninteracting systems [11]. The two relevant avoided crossing points are highlighted in the insets, where the energy gap between $|\Phi_1\rangle$ and $|\Phi_2\rangle$ for configuration II is 297 MHz while the gap between $|\Phi_3\rangle$ and $|\Phi_5\rangle$ for configuration III is 565 MHz. Note that the transition energies and entropies of the idle configuration I are included at $\lambda = 0$ (in both the left and right plots), while the dotted vertical lines indicate $\lambda = 1$, that is, the points where the system is kept during the hold time.

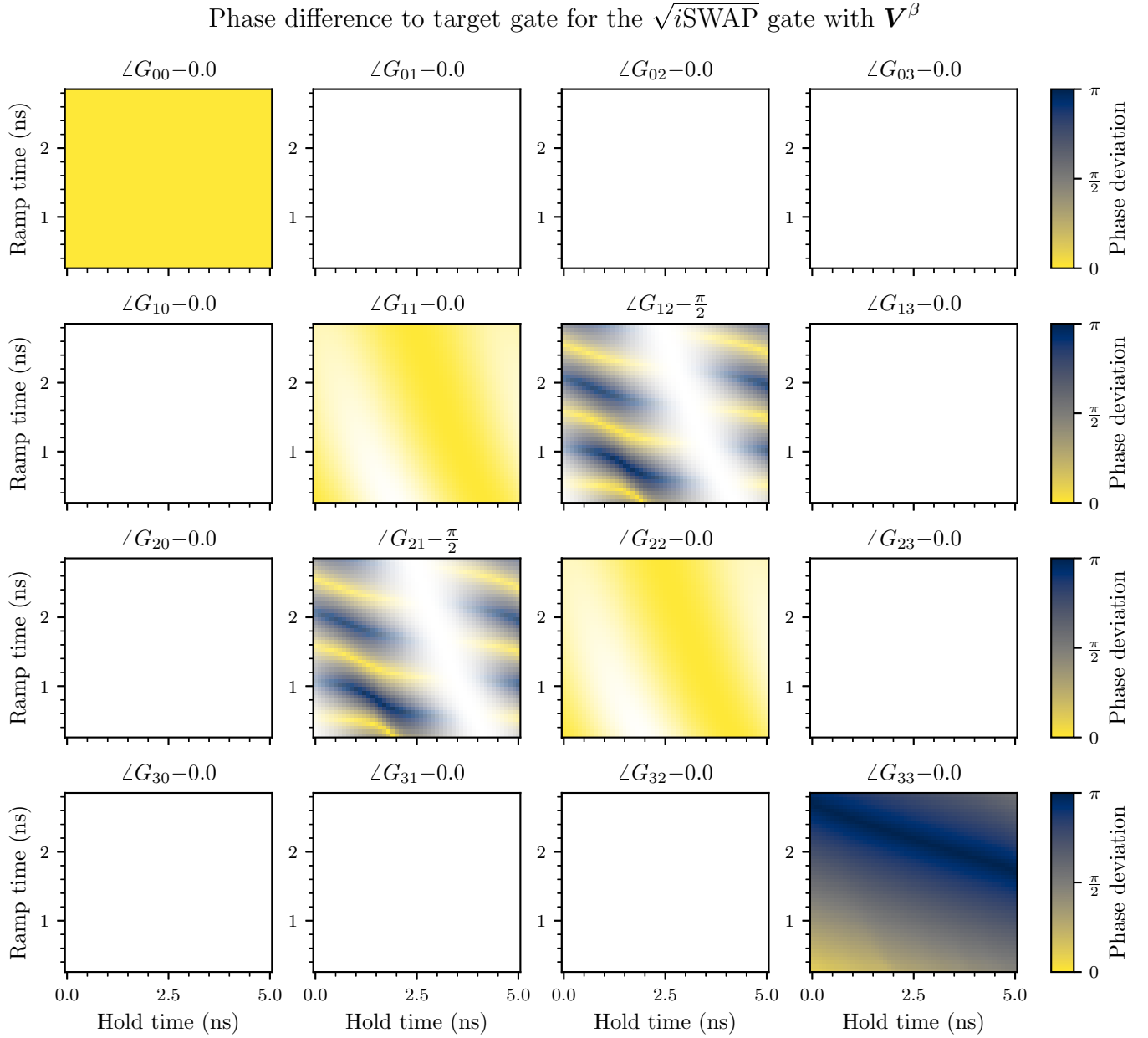


FIG. 7. Elementwise analysis of the $\mathbf{V}^\beta \sqrt{i}\text{SWAP}$ gate. Each plot corresponds to one element G_{ij} in the gate matrix $G(\theta_L, \theta_R)$, where the single-qubit rotation angles θ_L and θ_R have been chosen such that G_{ii} is real for $i \in \{0, 1, 2\}$. The color indicates the deviation from the ideal phase of the corresponding matrix element, as indicated in the respective plot titles, and the opacity signifies the amplitude. Note that the ideal amplitude, and thus opacity, of the four central plots is 0.5.

the single-qubit rotation angles according to

$$\theta_L = \angle U_{22} - \angle U_{00}, \quad \theta_R = \angle U_{11} - \angle U_{00}, \quad (\text{B1})$$

where $\angle U_{ii}$ denotes the complex phase of U_{ii} . This choice of angles collects the the phase of the diagonal elements of $G(\theta_L, \theta_R)$ on G_{33} such that G_{ii} is real for $i \in \{0, 1, 2\}$, thus allowing the general trends to emerge more clearly than they would with the optimized angles used in the main text. Throughout this Appendix, we adopt the notation G_{ij} for matrix element ij of the gate matrix $G(\theta_L, \theta_R)$ with θ_L and θ_R given by Eq. (B1). The phase deviation and amplitude of these matrix elements are displayed in Figs. 7 and 8 for the $\sqrt{i}\text{SWAP}$

gate and in Figs. 9 and 10 for the CZ gate, using $\mathbf{V}^\beta$ and $\mathbf{V}^\zeta$, respectively.

Figure 7 shows the elementwise analysis of the $\mathbf{V}^\beta \sqrt{i}\text{SWAP}$ gate. By comparing with the fidelity in Fig. 3(c), we see that the phase of the G_{33} element is limiting high gate fidelity to small values of both ramp and hold times. However, the amplitude of this matrix element shows no dependence on the gate parameters, indicating that interactions among higher excited states are suppressed, which is also illustrated in Fig. 11(a). Further structure in the fidelity stem from the phase and amplitude of the G_{12} and G_{21} elements.

The same data for the $\mathbf{V}^\zeta \sqrt{i}\text{SWAP}$ gate are illustrated in Fig. 8. We observe that the same matrix elements are

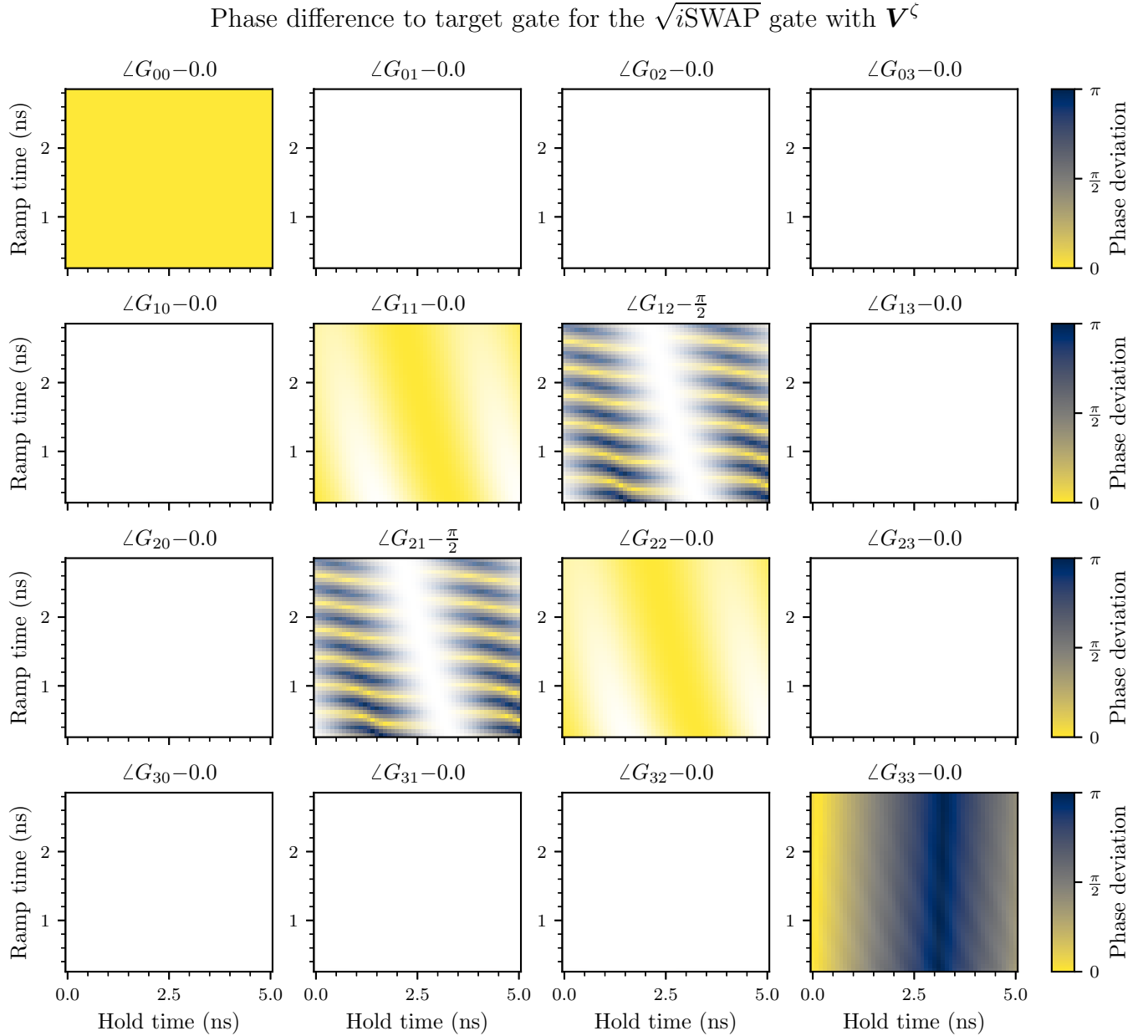


FIG. 8. Elementwise analysis of the $V^\zeta \sqrt{i\text{SWAP}}$ gate. Each plot corresponds to one element G_{ij} in the gate matrix $G(\theta_L, \theta_R)$, where the single-qubit rotation angles θ_L and θ_R have been chosen such that G_{ii} is real for $i \in \{0, 1, 2\}$. The color indicates the deviation from the ideal phase of the corresponding matrix element, as indicated in the respective plot titles, and the opacity signifies the amplitude. Note that the ideal amplitude, and thus opacity, of the four central plots is 0.5.

crucial with this voltage function. The G_{33} element is almost independent of the ramp time, likely because of the low ZZ coupling in configuration I, which is two orders of magnitude lower than in the V^β case. The small fluctuations in opacity are caused by interactions among the higher excited states, as shown in Fig. 11(b). The phases of the G_{12} and G_{21} elements prove to be crucial for the gate fidelity. The phase of both these elements oscillate with respect to the ramp time for both voltage functions, with a frequency of 1.04 GHz for V^β and 2.46 GHz for V^ζ . The frequency of these oscillations is closely related to the mean difference in energy between the $|\Psi_1(t)\rangle$ and $|\Psi_2(t)\rangle$ states during the ramp-up from configuration I to configuration II, which is 0.94 GHz for V^β and

2.51 GHz for V^ζ . Since the same energy difference during the hold and ramp-down stages of the gate operation is approximately 0.1 GHz and the phase is given by the integral of the energy over time, the phase difference of these matrix elements is dominated by the energy difference during the ramp-up stage.

For the CZ gates, illustrated in Fig. 9 for V^β and in Fig. 10 for V^ζ , the fidelity is evidently mainly determined by the G_{33} element. As discussed in the main text, the oscillations in amplitude of this matrix element stem from population leakage from $|11\rangle$ to the nonqubit states $|02\rangle$ and $|20\rangle$. Figures 11(c) and 11(d) show these oscillations as a function of hold time for V^β and V^ζ , respectively. For the full fidelity

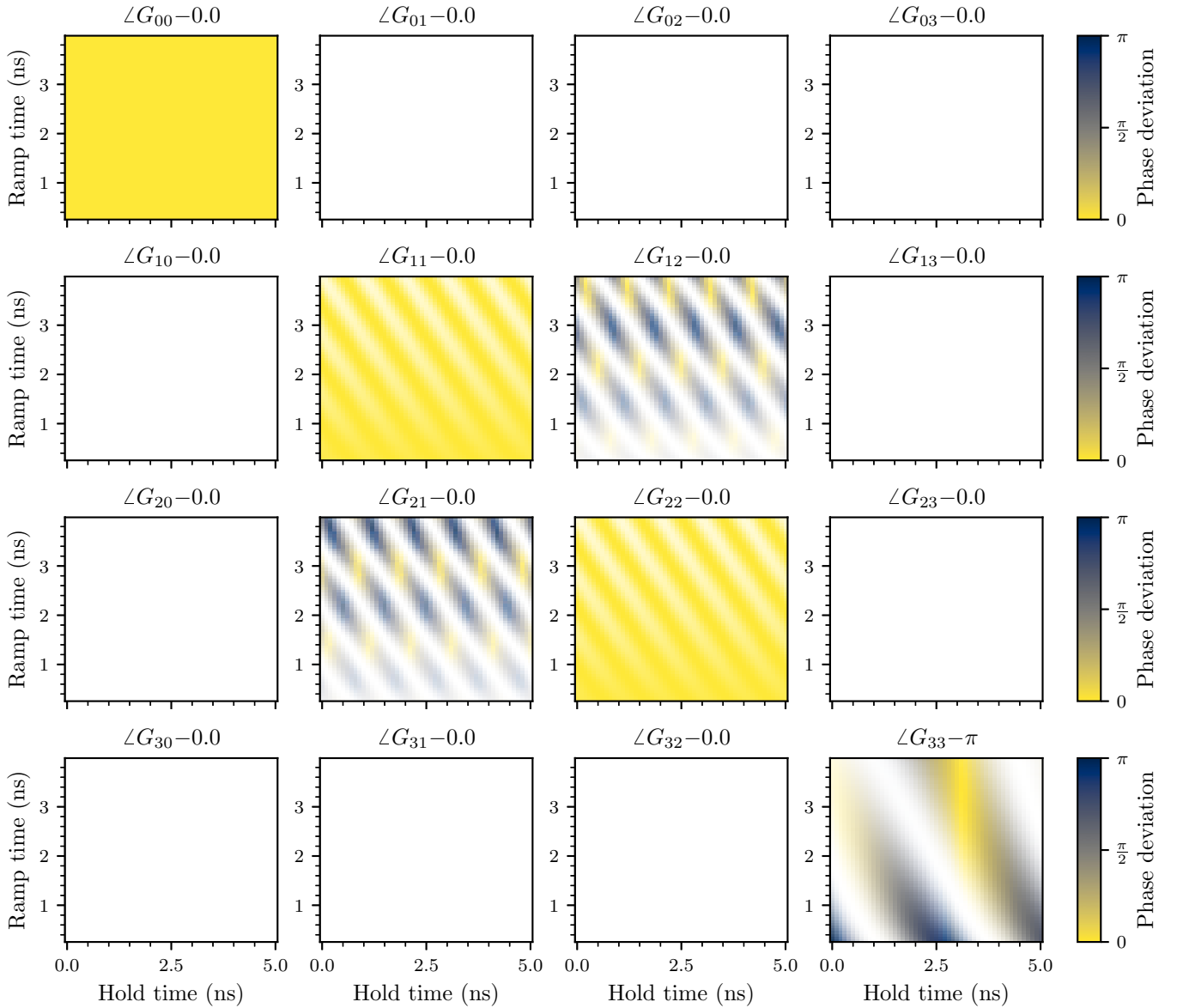
Phase difference to target gate for the CZ gate with V^β 

FIG. 9. Elementwise analysis of the V^β CZ gate. Each plot corresponds to one element G_{ij} in the gate matrix $G(\theta_L, \theta_R)$, where the single-qubit rotation angles θ_L and θ_R have been chosen such that G_{ii} is real for $i \in \{0, 1, 2\}$. The color indicates the deviation from the ideal phase of the corresponding matrix element, as indicated in the respective plot titles, and the opacity signifies the amplitude. The off-diagonal plots should ideally have opacity 0, i.e., be white.

of V^β , some additional complexity is added by the central elements, i.e., G_{ij} with $i, j \in \{1, 2\}$. This occurs because of an unwanted interaction between the $|\Psi_1(t)\rangle$ and $|\Psi_2(t)\rangle$ states, which stems from the passage through configuration II when ramping the V^β function to configuration III. This unwanted interaction is essentially eliminated with the V^ζ function since the parametrizations for the different gates are completely separated.

APPENDIX C: FULL TIME EVOLUTION

As stated in Sec. II, the computational basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|02\rangle$, $|11\rangle$, and $|20\rangle$ are defined by the six lowest eigenstates $|\Phi_n\rangle$ at $t = 0$, which is when the system

is in configuration I [11]. Among them, the four relevant qubit states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, i.e., the eigenstates $|\Phi_0\rangle$, $|\Phi_1\rangle$, $|\Phi_2\rangle$, and $|\Phi_4\rangle$, were propagated in time. In this Appendix, we present the overlap $|\langle\Psi_i(t)|\Phi_n\rangle|^2$ between the four propagated qubit states $|\Psi_i(t)\rangle$, $i \in \{0, 1, 2, 4\}$, where $|\Psi_n(0)\rangle = |\Phi_n\rangle$, and the six computational basis states $|\Phi_n\rangle$, with $n \in \{0, \dots, 5\}$, through the whole time evolution from $t = 0$ to $t = t_{\text{gate}}$. Figure 12 illustrates the dynamics for the $\sqrt{i}$ SWAP gates. We observe that all states become superpositions of the computational basis states when the system is in the interacting stage of the gate. This is not necessarily because of a population transfer, but rather because the configuration I eigenfunctions do not constitute an

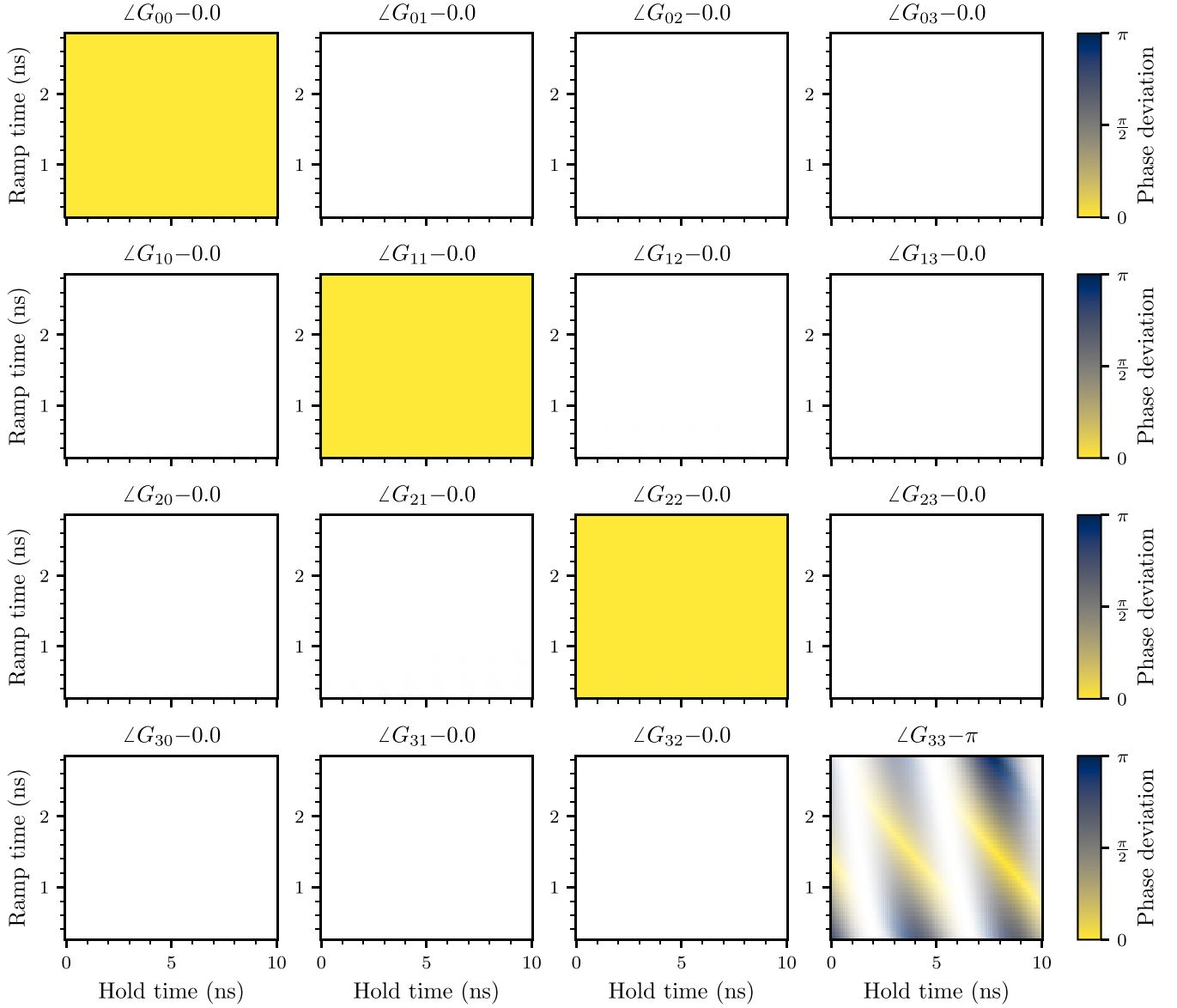
Phase difference to target gate for the CZ gate with V^ζ 

FIG. 10. Elementwise analysis of the V^ζ CZ gate. Each plot corresponds to one element G_{ij} in the gate matrix $G(\theta_L, \theta_R)$, where the single-qubit rotation angles θ_L and θ_R have been chosen such that G_{ii} is real for $i \in \{0, 1, 2\}$. The color indicates the deviation from the ideal phase of the corresponding matrix element, as indicated in the respective plot titles, and the opacity signifies the amplitude.

eigenbasis for configuration II. A completely adiabatic transition thus requires the eigenstates to become superpositions in the configuration I eigenbasis. Consequently, the $|\Psi_0(t)\rangle$ states might at first glance appear to interact with other states, but the completely static composition during the hold stage of the gate reveals that this is not the case. The same is true for $|\Psi_4(t)\rangle$ for V^β , whereas small unwanted oscillations are observed for the same state in the case of V^ζ . These oscillations are also visible in the G_{33} plot of Fig. 8. They are however compensated for by an appropriate choice of hold and ramp times.

In the case of the CZ gates, illustrated in Fig. 13, the static behavior discussed above is desired for $0 \leq i \leq 2$. This ideal behavior is observed for V^ζ , as seen in Figs. 13(e)–13(g),

whereas V^β is subject to unwanted oscillations of the $|\Psi_1(t)\rangle$ and $|\Psi_2(t)\rangle$ states, illustrated in Figs. 13(b) and 13(c), respectively. As discussed in the main text and in Appendix B, this interaction is responsible for the sensitivity to ramp time observed for the V^β CZ gate.

APPENDIX D: SCREENING OF THE COULOMB INTERACTION

In order to estimate the impact of screening by the electrodes beneath the helium layer on gate time, we approximate the seven electrodes as an infinite metallic surface. Note that this is a significant simplification, as the empty spaces between the electrodes in the actual device are as wide as the

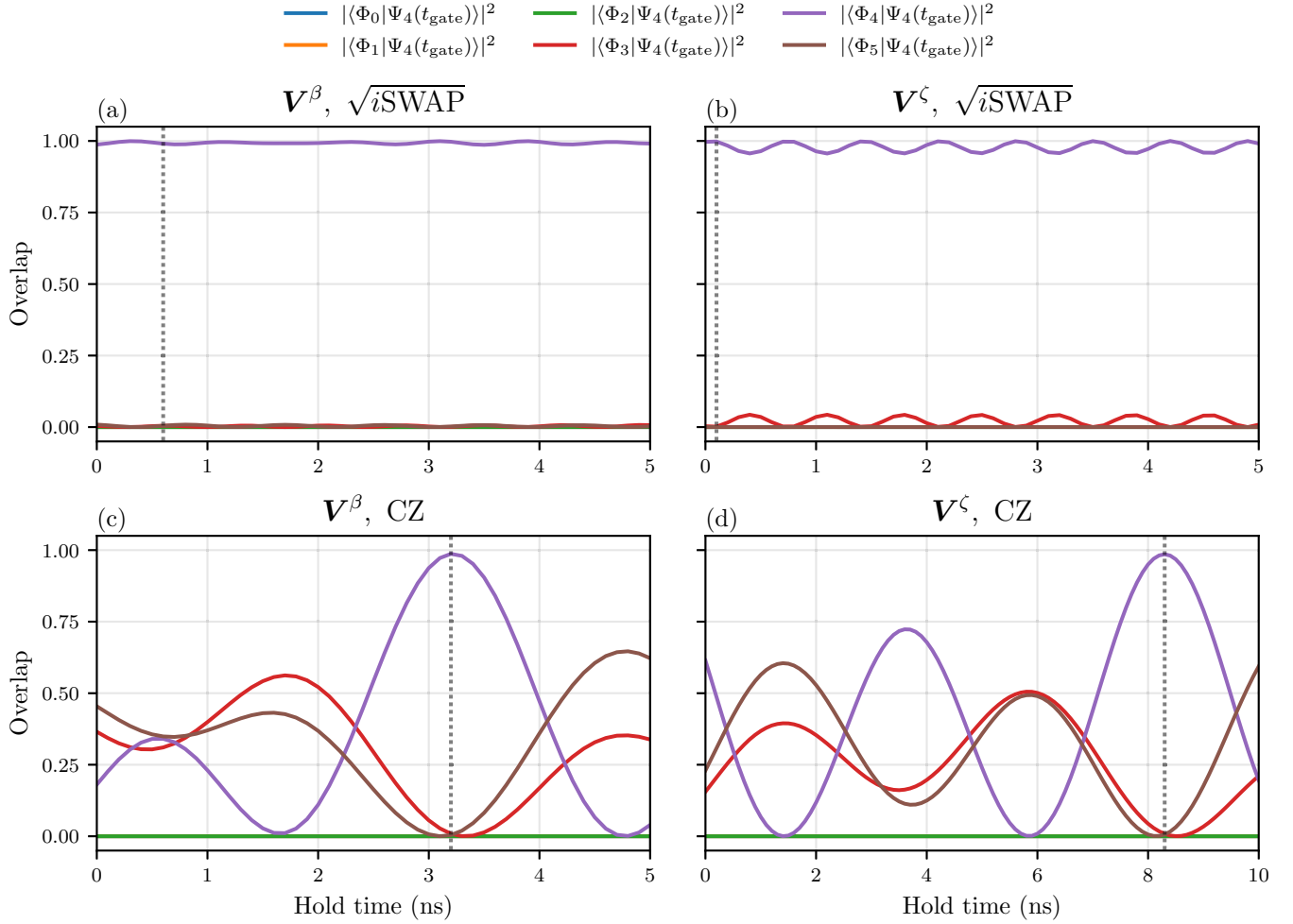


FIG. 11. Overlap between the state $|\Psi_4(t_{\text{gate}})\rangle$, propagated by different two-qubit gate protocols, and the eigenstates $|\Phi_n\rangle$ before the gate protocol starts, as a function of hold time. All plots use the optimal ramp time for the respective gates and the optimal hold times are indicated by dotted vertical lines. The red and brown lines correspond to leakage from $|11\rangle$ to $|02\rangle$ and $|20\rangle$, respectively. For an ideal, leakage free, gate the overlap $|\langle\Phi_4|\Psi_4(t_{\text{gate}})\rangle|^2$ (purple line) should be 1.

electrodes themselves. This allows us to calculate the screened Coulomb interaction via the method of image charges [12]. Since the Coulomb interaction depends solely on the relative distance between the electrons, we express the interaction as a function of the separation $r = |x_1 - x_2|$, denoted by $u(r)$. The resulting interaction, including both the direct repulsion between the electrons and the attraction between each electron and the image charge of the other, can be expressed as

$$u(r) = \frac{\kappa}{r} - \frac{\kappa}{\sqrt{r^2 + (2h)^2}}, \quad (\text{D1})$$

where h is the distance between the electrons and the conducting surface and κ is the Coulomb strength parameter defined in Eq. (1).

To determine the impact of screening on the gate speed, we perform a Taylor expansion of the interaction around the equilibrium electron separation d . Writing the instantaneous separation as $r = d + (\Delta x_2 - \Delta x_1)$, where Δx_i is a small displacement of electron i , the expansion to second order

yields

$$u(r) \approx u(d) + u'(d)(\Delta x_2 - \Delta x_1) + \frac{1}{2}u''(d)(\Delta x_2 - \Delta x_1)^2. \quad (\text{D2})$$

We note that the static term $u(d)$ and the force terms proportional to $u'(d)$ only lead to shifts in the total energy and equilibrium positions, as they consist solely of single-particle terms where the electron displacements appear independently. While significant, these static shifts primarily affect the qubit frequencies and the resonance conditions for the gates. In our protocol, these effects can be compensated for by modifying the external potential through reoptimization of the voltage functions $V(\lambda)$ to restore the necessary avoided crossings between energy levels.

Expanding the quadratic term reveals the operator $-u''(d)\Delta x_1\Delta x_2$. This cross term is the only term in the expansion that couples the two electrons. Consequently, it is this term that drives the gate dynamics and determines the gate speed. We therefore focus on its strength $u''(d)$. To quantify it, we define the screening factor η as the ratio of the screened

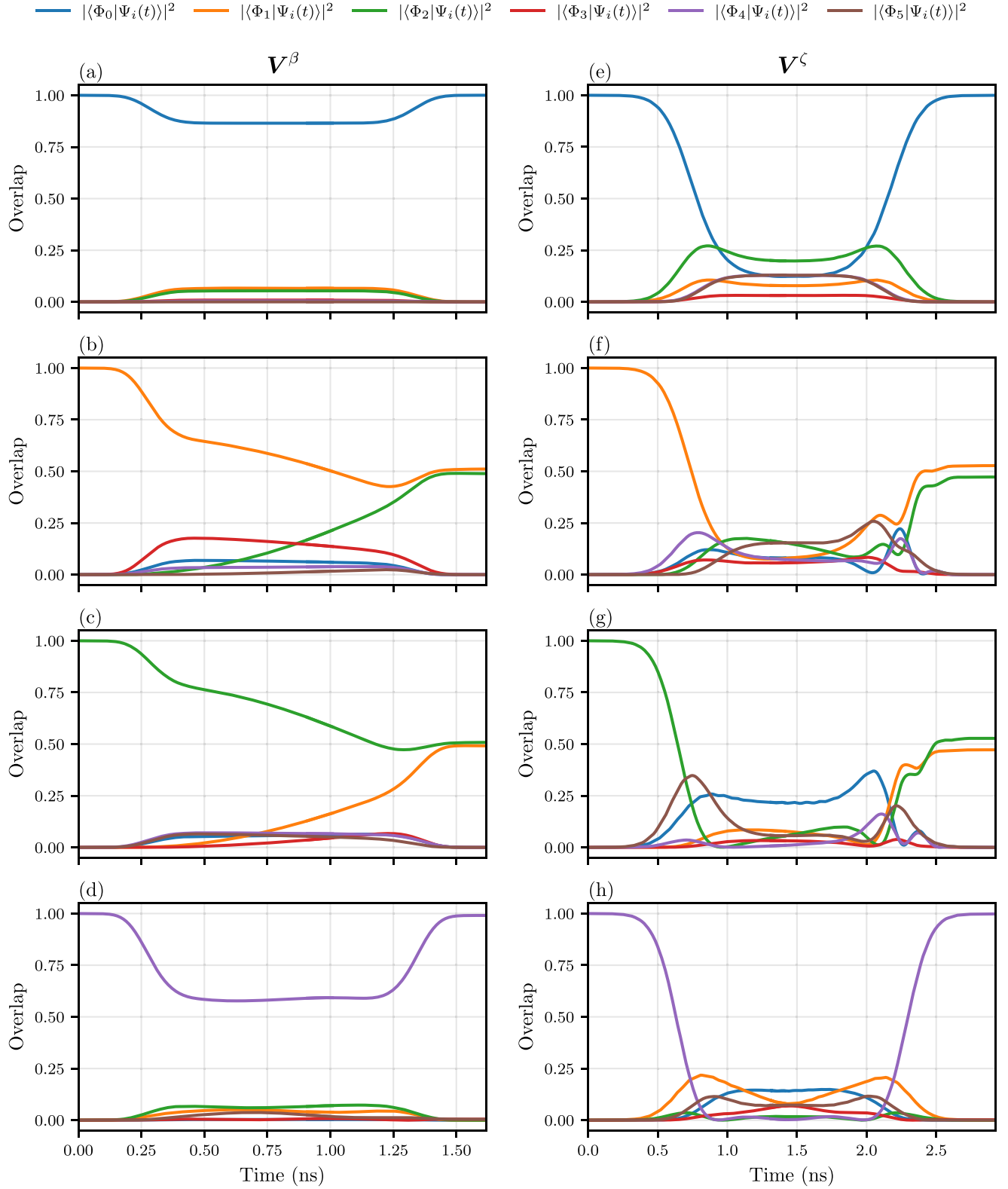


FIG. 12. Time evolution of the overlaps between state $|\Psi_i(t)\rangle$ and the six lowest-energy eigenstates at time $t = 0$ during the optimal operation of the $\sqrt{i\text{SWAP}}$ gate using (a)–(d) V^β and (e)–(h) V^ζ . Different plots show different states $|\Psi_i(t)\rangle$ for (a) and (e) $i = 0$, (b) and (f) $i = 1$, (c) and (g) $i = 2$, and (d) and (h) $i = 4$.

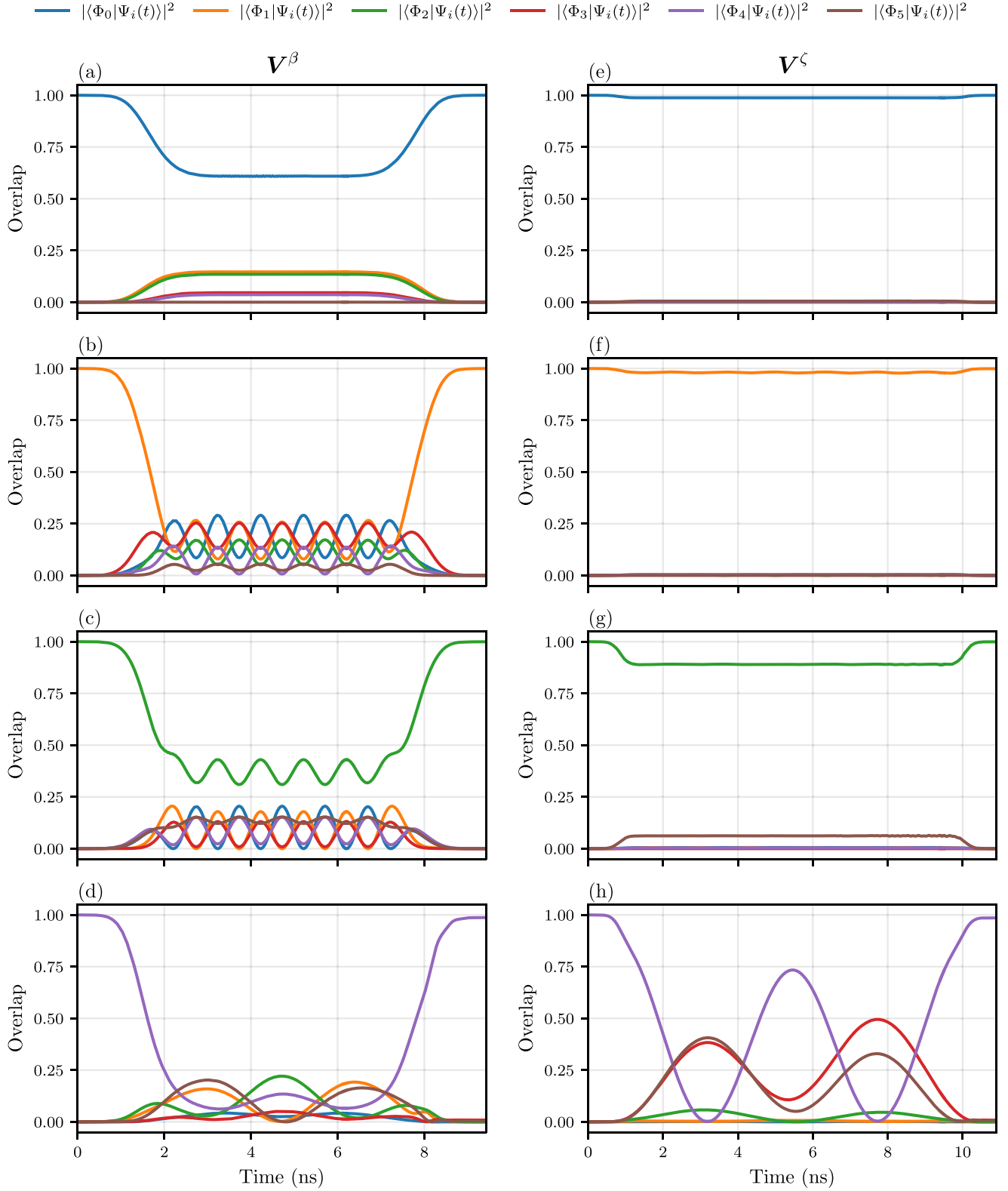


FIG. 13. Time evolution of the overlaps between state $|\Psi_i(t)\rangle$ and the six lowest-energy eigenstates at time $t = 0$ during the optimal operation of the CZ gate using (a)–(d) V^β and (e)–(h) V^ζ . Different plots show different states $|\Psi_i(t)\rangle$ for (a) and (e) $i = 0$, (b) and (f) $i = 1$, (c) and (g) $i = 2$, and (d) and (h) $i = 4$.

curvature to the unscreened curvature. The unscreened direct interaction $u_{\text{dir}}(r) = \kappa/r$ has a curvature $u''_{\text{dir}}(d) = 2\kappa/d^3$. For the image interaction term $u_{\text{img}}(r) = -\kappa(r^2 + 4h^2)^{-1/2}$, the second derivative at $r = d$ is

$$u''_{\text{img}}(d) = -2\kappa \frac{d^2 - 2h^2}{(d^2 + 4h^2)^{5/2}}. \quad (\text{D3})$$

The screening factor is then given by

$$\eta = \frac{u''_{\text{dir}}(d) + u''_{\text{img}}(d)}{u''_{\text{dir}}(d)} = 1 - \frac{d^3(d^2 - 2h^2)}{(d^2 + 4h^2)^{5/2}}. \quad (\text{D4})$$

Defining the ratio $\xi = 2h/d$, this simplifies to

$$\eta = 1 - \frac{1 - \xi^2/2}{(1 + \xi^2)^{5/2}}. \quad (\text{D5})$$

For our device geometry, with $d = 1.5 \mu\text{m}$ and $h = 500 \text{ nm}$, the expression above yields a screening factor of $\eta \approx 0.69$.

APPENDIX E: ELECTRON-RIPPLON COUPLING HAMILTONIAN

The electron coupling to the ripplon bath follows directly from the interaction Hamiltonian

$$\hat{\mathcal{H}}_i = \sum_{\mathbf{q}} V_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} (b_{\mathbf{q}} + b_{-\mathbf{q}}^\dagger), \quad (\text{E1})$$

where $b_{\mathbf{q}}$ is the annihilation operator of a ripplon with wave vector $\mathbf{q}$, $\mathbf{r} = (x, y)$ is the electron coordinate, and $V_{\mathbf{q}}$ is the electron coupling parameter to the ripplonic field [9,10]. The coupling parameter is determined by a pressing field $E_\perp$, which is a controlled parameter in the experimental devices,

and a polarization term $\mathcal{R}(q)$ [41],

$$V_{\mathbf{q}} = \sqrt{\frac{\hbar q}{2\rho\omega_{\text{q}}S_{\text{He}}}} [eE_\perp + \mathcal{R}(q)]. \quad (\text{E2})$$

Here $\omega_{\mathbf{q}} = \sqrt{\sigma_{\text{He}}/\rho_{\text{He}}q^3}$ is the ripplon dispersion relation, σ_{He} is the surface tension of liquid helium, ρ_{He} is the liquid-helium density, and S_{He} is the helium surface area. The dominant contribution to the interaction Hamiltonian arises from ripples with wave vectors q_x near the inverse electron localization length $l_x^{-1} = \sqrt{\hbar/m_e\omega_e}$, which is approximately 40 nm for electron motional frequencies of $\omega_e/2\pi = 10 \text{ GHz}$. Typical frequencies of ripples at these wave vectors, $\omega_{\mathbf{q}}/2\pi \approx 0.4 \text{ GHz}$, are small compared to ω_e . Therefore, single-ripplon electron motional state decay processes are strongly suppressed. However, ripples can strongly contribute to dephasing by inducing fluctuations in the energies of the motional states [32,33]. In the adiabatic approximation the energy modulation strength is characterized by a dimensionless parameter

$$\mathcal{C} = \sum_{\mathbf{q}} |\alpha_{\mathbf{q}}|^2 (2\bar{n}_{\mathbf{q}} + 1), \quad (\text{E3})$$

where $\alpha_{\mathbf{q}} \approx V_{\mathbf{q}}/\hbar\omega_{\mathbf{q}}$ for relevant parameters $q_x l_x \sim 1$ and $\bar{n}_{\mathbf{q}}$ is the thermal occupation number of a ripplon with the wave number $\mathbf{q}$. The absorption spectrum linewidth γ in the strong-coupling regime is then given by

$$\gamma = \left(\sum_{\mathbf{q}} |\alpha_{\mathbf{q}}|^2 (2\bar{n}_{\mathbf{q}} + 1) \omega_{\mathbf{q}}^2 \right)^{1/2}. \quad (\text{E4})$$

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